

HisabDo AI — Day 14

AI Budget Recommendation — Team Lead (Taha)

1. Purpose

Define the MVP requirements and explainable logic for recommending category-level budgets from verified HisabDo financial data. Backend calculations are the source of truth; AI only explains verified results.

2. Taha — Exact Task Scope

1. Define and finalize Day 14 requirements.
2. Define budget recommendation objectives.
3. Define required income, expense, category and existing budget data.
4. Finalize budget recommendation logic.
5. Define category-wise budget recommendation rules.
6. Define how AI should use verified backend calculations.
7. Define recommendation output structure.
8. Define insufficient-data and missing-budget handling.
9. Coordinate Jaffer, Omesha, Aarti and Laiba.
10. Review implementation and testing.
11. Validate recommendations against sample financial data.
12. Prepare Day 14 documentation and acceptance checklist.

3. Budget Recommendation Objectives

- Help users create realistic category-level budgets.
- Use historical spending instead of arbitrary amounts.
- Identify categories where verified spending is above the existing budget.
- Avoid strong recommendations when data is insufficient.
- Keep recommendations explainable and actionable.
- Make every numeric recommendation traceable to backend calculations.

4. Required Data

- User ID
- Transaction ID
- Transaction Type
- Amount
- Category
- Transaction Date
- Existing Budget Amount (where available)

Optional: recurring expenses and historical budgets. Actual HisabDo schema must be mapped before production.

5. MVP Budget Recommendation Logic

1. Use the latest 3–6 complete months where available.
2. Calculate monthly expense totals for each category.
3. Calculate the average monthly spending for each category.
4. Use the historical average as the MVP recommended-budget baseline.
5. Apply only approved/configurable planning rules; do not invent arbitrary values.
6. Compare the recommendation with the existing budget when available.

6. Category-wise Rules

Normal category: Recommended budget uses the verified historical monthly average.

Existing budget below historical spending: Flag the category for review because spending is above the current budget.

Existing budget above historical spending: Show the comparison; do not force a reduction.

No existing budget: Recommend from sufficient historical spending data.

Insufficient category history: Mark as Limited Data and avoid strong conclusions.

Zero/negative income: Do not make affordability claims without verified income data.

7. Calculation Definitions

- Category Monthly Average = Total category expense ÷ number of months in the analysis period.
- Budget Variance = Existing Budget – Historical Average.
- Budget Utilization = Historical Average ÷ Existing Budget × 100, when existing budget > 0.
- Recommended Budget = verified historical baseline under the configured MVP rules.

8. AI Usage / No-Invention Rules

- AI receives only verified recommendation results.
- AI explains why a category received its recommendation.
- AI explains current vs recommended budget when supplied.
- AI can provide practical observations based on verified results.
- AI must never invent, change, or recalculate a financial amount.
- AI must not claim affordability without verified supporting data.
- Missing values must be reported as unavailable, not guessed.

9. Recommendation Output Structure

- UserId
- AnalysisPeriod
- Category
- HistoricalAverage
- ExistingBudget
- RecommendedBudget
- BudgetVariance
- BudgetUtilization
- RecommendationReason
- DataSufficiency
- Limitations

10. Missing / Insufficient Data Handling

- No transactions → no numeric recommendation.
- Insufficient history → Limited Data status.
- No existing budget → recommendation is still allowed if history is sufficient.
- Missing category → exclude or flag; never guess a category.
- Existing budget = 0 → utilization is unavailable; avoid division by zero.
- Income unavailable → no affordability conclusion.

11. Sample Verified Result

Food | Historical Average: PKR 15,000 | Existing Budget: PKR 12,000 | Recommended Budget: PKR 15,000 | Reason: verified recent spending is above the current budget.

This is a testing example only. Production values must come from HisabDo.

12. Sample AI Explanation

Your recent verified spending in Food averages PKR 15,000 per month, while your current budget is PKR 12,000. The recommended budget is PKR 15,000 based on the available historical spending

pattern. You can review this category if you want to reduce spending, but the recommendation is based on historical behavior and is not a guaranteed future expense.

13. Architecture

HisabDo Database → Budget Calculation/Recommendation Engine → Verified Recommendation Result → AI Explanation Layer → Mobile/Web UI

14. Coordination & Review

- Jaffer: backend data, calculations, API, authorization.
- Omesha: recommendation methodology, category logic, validation.
- Aarti: AI explanation and no-invention enforcement.
- Laiba: API/UI integration.
- Taha: requirements, architecture, coordination, review, validation and acceptance.

15. Testing & Validation

- Normal category
- Existing budget below spending
- Existing budget above spending
- No existing budget
- New category
- Insufficient history
- No transactions
- Zero budget
- Missing category
- Missing income
- Multiple-user authorization
- AI output matches backend values exactly

16. Acceptance Checklist

- ☐ Define and finalize Day 14 requirements.
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- ☐ Define how AI should use verified backend calculations.

- ☐ Define recommendation output structure.
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17. Final Definition

Day 14 provides category-level budget guidance from verified HisabDo historical spending and existing budgets. The MVP uses a simple historical baseline. Backend generates numeric recommendations; AI explains them without inventing financial values.